

```
import pandas as pd
Inport numpy as np
df = pd.read_csv("/content/sample_data/california_housing_test.csv")
```

```
df.head()
df.tail()
```

	longitude	latitude	housin	median_age	total_rooms	total_bedrooms	population	households	median_income	median_house_
2995	-119.86	34.42		23.0	1450.0	642.0	1258.0	607.0	1.1790	225
2996	-118.14	34.06		27.0	5257.0	1082.0	3496.0	1036.0	3.3906	237
2997	-119.70	36.30		10.0	956.0	201.0	693.0	220.0	2.2895	62
2998	-117.12	34.10		40.0	96.0	14.0	46.0	14.0	3.2708	162
2999	-119.63	34.42		42.0	1765.0	263.0	753.0	260.0	8.5608	500

```
df.shape
```

```
(3000, 9)
```

```
df.columns
```

```
Index(['longitude', 'latitude', 'housing_median_age', 'total_rooms',
       'total_bedrooms', 'population', 'households', 'median_income',
       'median_house_value'],
      dtype='object')
```

```
df.dtypes
```

	0
longitude	float64
latitude	float64
housing_median_age	float64
total_rooms	float64
total_bedrooms	float64
population	float64
households	float64
median_income	float64
median_house_value	float64

```
dtype: object
```

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 3000 entries, 0 to 2999
Data columns (total 9 columns):
#   Column                Non-Null Count  Dtype
0   longitude             3000 non-null  float64
1   latitude              3000 non-null  float64
2   housing_median_age    3000 non-null  float64
3   total_rooms           3000 non-null  float64
4   total_bedrooms        3000 non-null  float64
5   population            3000 non-null  float64
6   households            3000 non-null  float64
7   median_income         3000 non-null  float64
8   median_house_value    3000 non-null  float64
dtypes: float64(9)
memory usage: 211.1 KB
```

```
df.isnull().sum()
df.isnull().values.any()
```

```
np.False_
```

```
(df == 0).sum()
df.select_dtypes(include=np.number).eq(0).sum()
```

```

      0
longitude  0
latitude   0
housing_median_age  0
total_rooms   0
total_bedrooms  0
population   0
households   0
median_income  0
median_house_value  0

```

```
dtype: int64
```

```
df.describe(include='all')
```

	longitude	latitude	housing_median_age	total_rooms	total_bedrooms	population	households	median_income	median_
count	3000.000000	3000.000000	3000.000000	3000.000000	3000.000000	3000.000000	3000.000000	3000.000000	
mean	-119.589200	35.63539	28.845333	2599.578667	529.950667	1402.798667	489.91200	3.807272	2
std	1.994936	2.12967	12.555396	2155.593332	415.654368	1030.543012	365.42271	1.854512	
min	-124.180000	32.56000	1.000000	6.000000	2.000000	5.000000	2.00000	0.499900	
25%	-121.810000	33.93000	18.000000	1401.000000	291.000000	780.000000	273.00000	2.544000	1
50%	-118.485000	34.27000	29.000000	2106.000000	437.000000	1155.000000	409.50000	3.487150	1
75%	-118.020000	37.69000	37.000000	3129.000000	636.000000	1742.750000	597.25000	4.656475	2
max	-114.490000	41.92000	52.000000	30450.000000	5419.000000	11935.000000	4930.00000	15.000100	

```
df[['longitude', 'latitude']]
```

	longitude	latitude
0	-122.05	37.37
1	-118.30	34.26
2	-117.81	33.78
3	-118.36	33.82
4	-119.67	36.33
2995	-119.86	34.42
2996	-118.14	34.06
2997	-119.70	36.30
2998	-117.12	34.10
2999	-119.63	34.42

```
3000 rows 2 columns
```

```
df['latitude'] = df['latitude'].astype(float)
df['latitude'] = df['latitude'].astype(int)
```

```
np.mean(df['median_house_value'])
DD.Max(df['median_house_value'])
```

```
np.min(df[ 'median_house_value' ])
```

```
22500.0
```